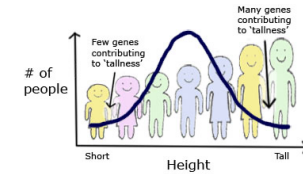


Module – Complex Trait (Quantitative) Genetics



Lecture 13— When many genes and many environmental effects influence trait expression and variation

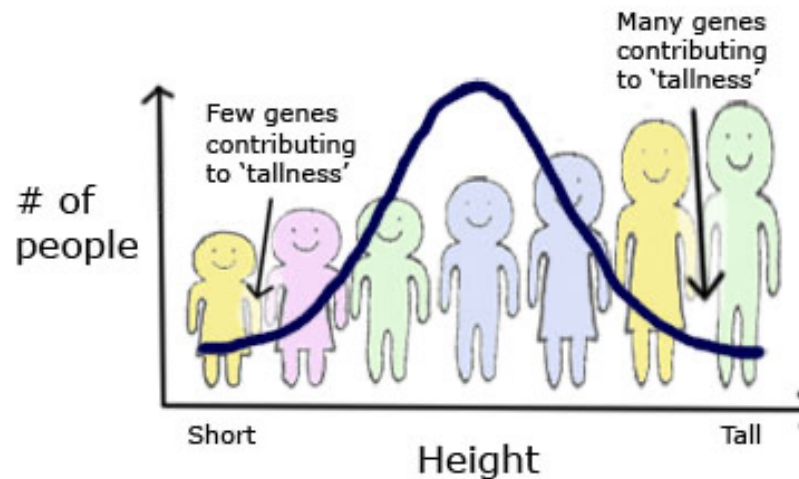
1. Complex trait variation (**Ch19 Section 1**)
2. Measuring components of phenotypic trait variation (**Ch19 Sections 2,3**)
3. Quantitative trait loci: Using molecular markers to infer the presence of genes affecting genetic variance (**Ch19 reading**)

PRIMER—Quantitative Genetics (see myCourses page for Biology 202)

Self-Assessment Exercise 2: Wed, 5 February at 08h55-09h25

Module: Complex trait genetics

Lecture 13— When many genes and many environmental effects influence trait expression and variation

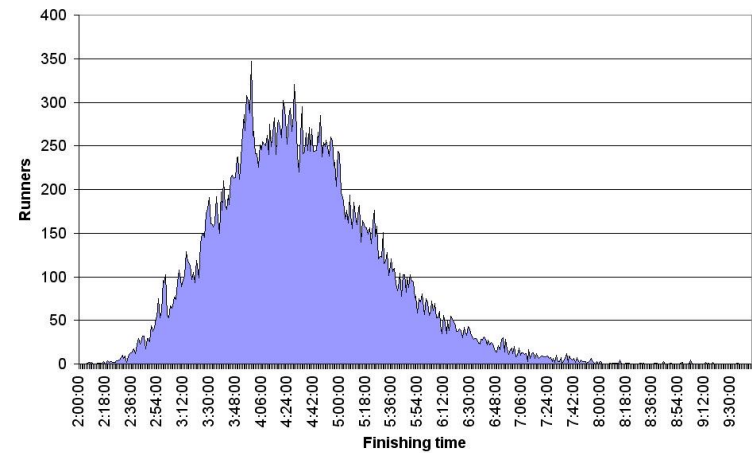


Types of complex traits

Continuous



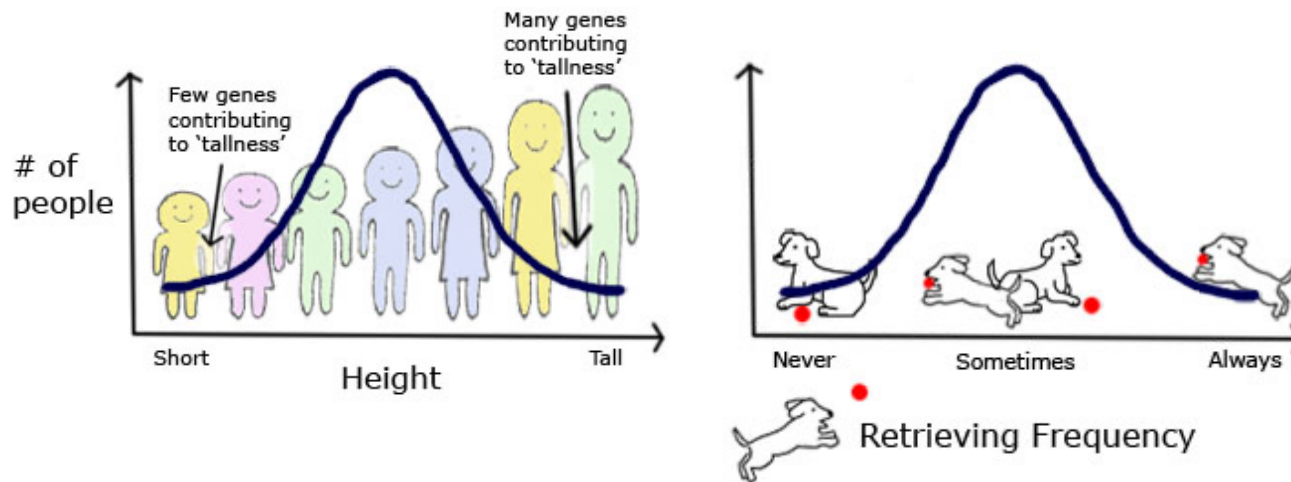
Body size (height)

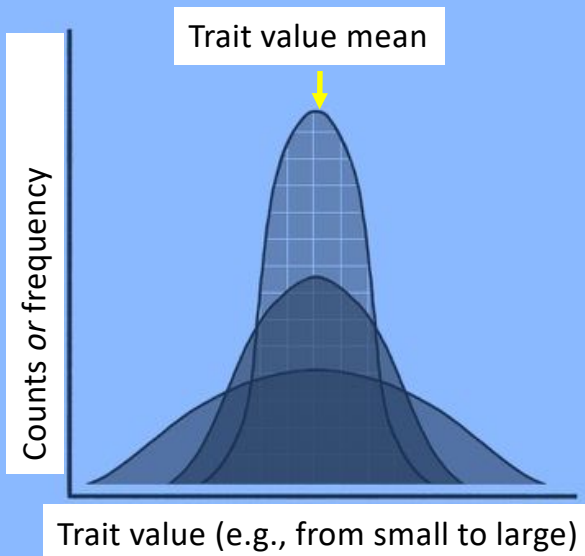


*Running speed
(e.g., finishing time in a marathon)*

Complex (quantitative) trait genetics

The nature of quantitative variation:
Distribution of phenotypic states is often “bell-shaped”





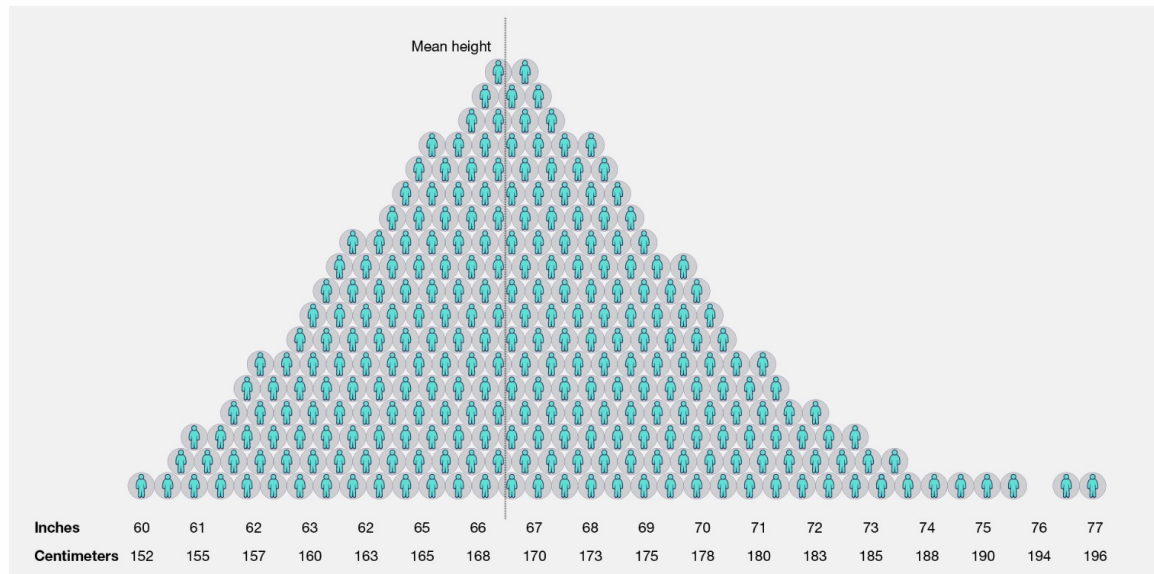
Variance

[ˈver-ē-ən(t)s]

A measurement of how far each number in a data set is from the mean (average), and thus from every other number in the set.

When we look at a population, we can measure the PHENOTYPIC variance

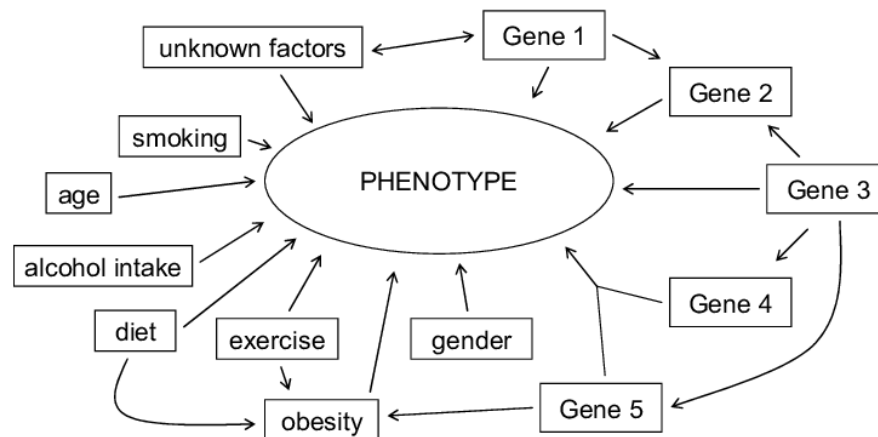
Phenotypic variance is the observed variability of a particular trait. For many measurable traits (such as height, skin color, and blood pressure) phenotypic variation can be thought of as a distribution. The variation within that distribution centers on an average or mean. Phenotypic variance is often caused by a combination of genetic and environmental factors.



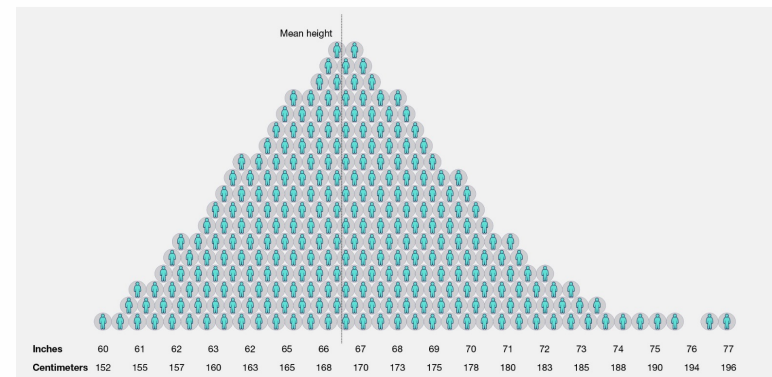
The “Multifactorial Model” for complex trait variation

ENVIRONMENTAL FACTORS

GENETIC FACTORS

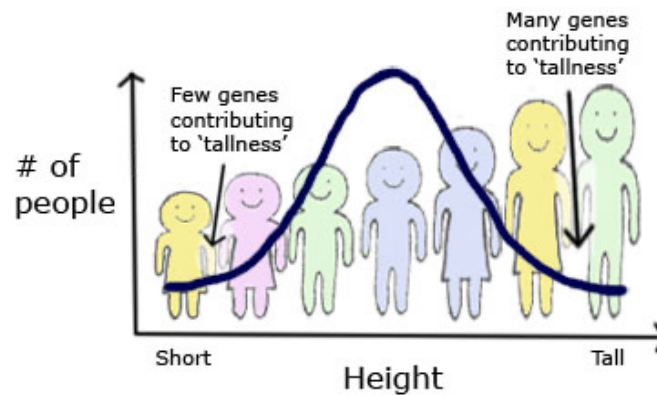


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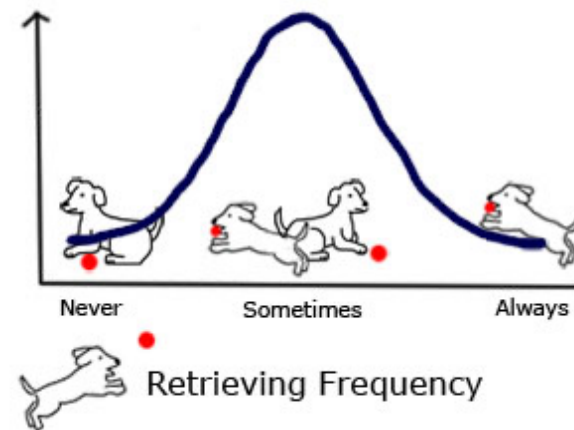


Why do complex traits usually show bell-shaped distributions?

Phenotypic variance of height



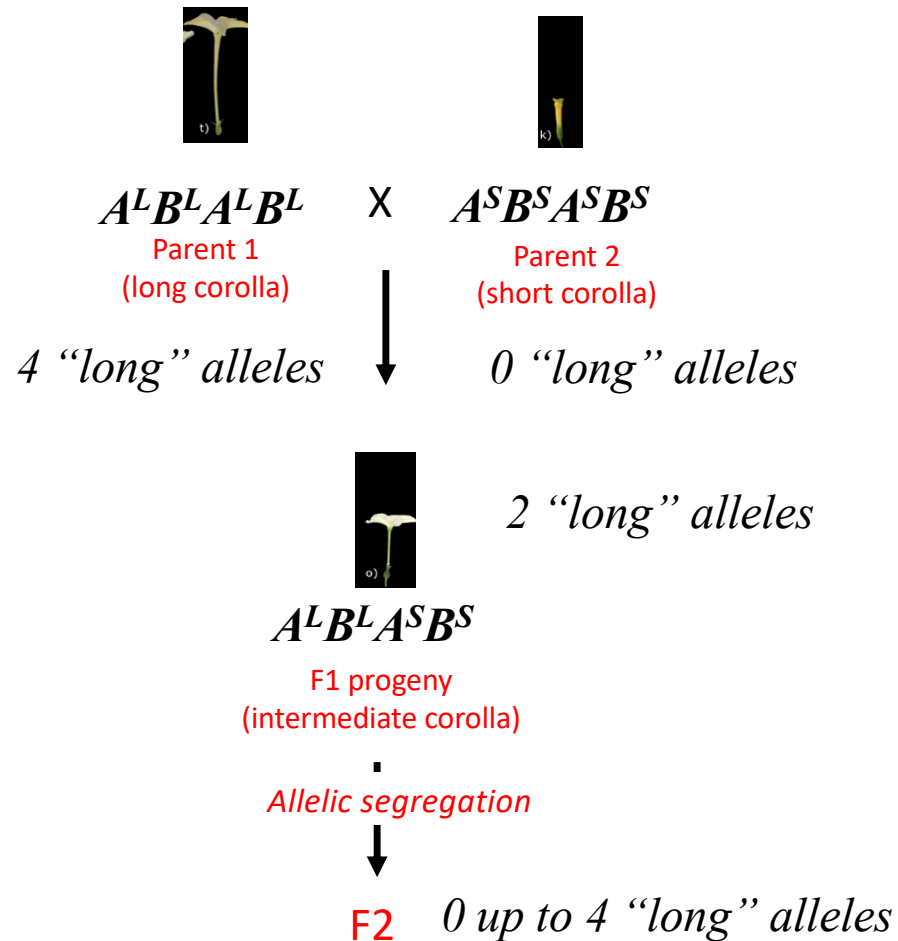
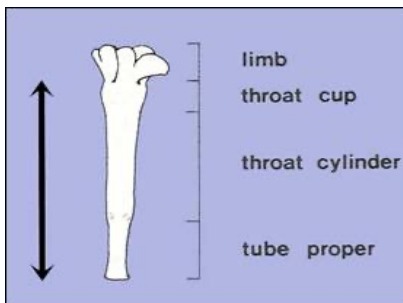
Phenotypic variance of retrieving frequency



Why do complex traits usually show bell-shaped distributions?



Tobacco

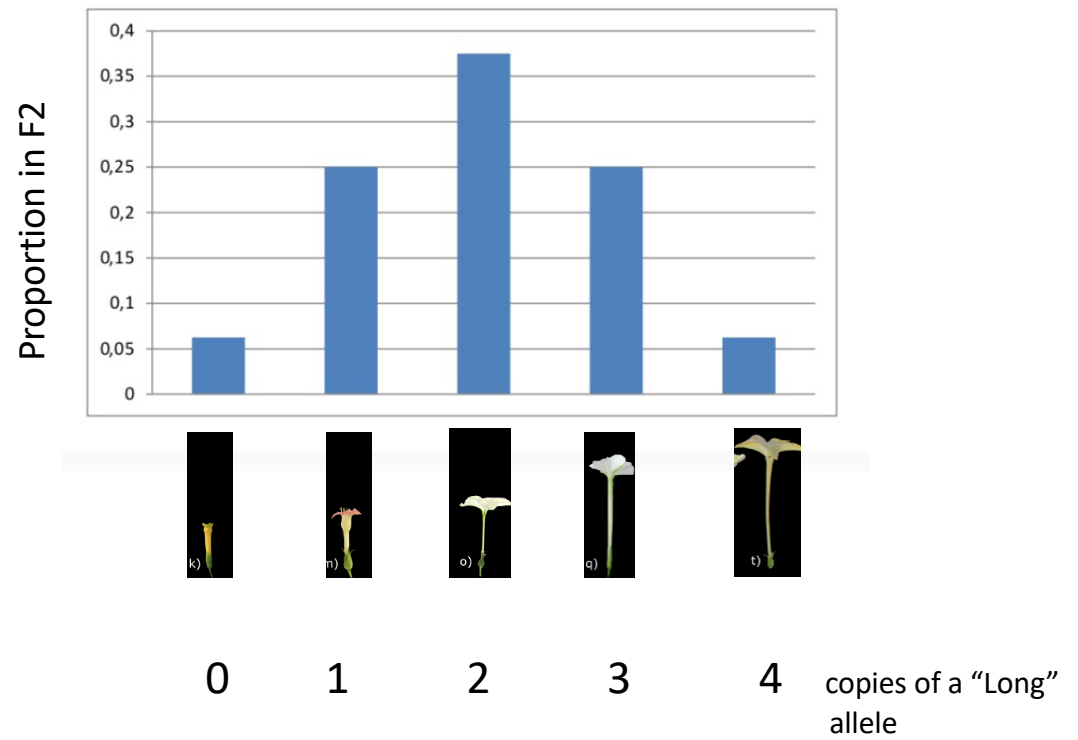


F2 generation

$$A^L B^L A^S B^S \times A^L B^L A^S B^S$$

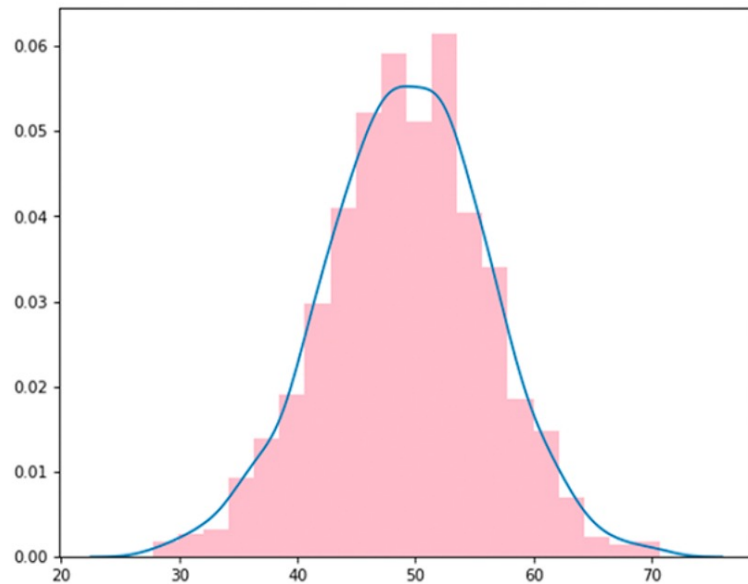
	$A^L B^L$	$A^L B^S$	$A^S B^L$	$A^S B^S$
$A^L B^L$	4	3	3	2
$A^L B^S$	3	2	2	1
$A^S B^L$	3	2	2	1
$A^S B^S$	2	1	1	0

Number of “Long” alleles



Why do complex traits usually show bell-shaped distributions?

(apart from the genetics, there are often many environmental effects)

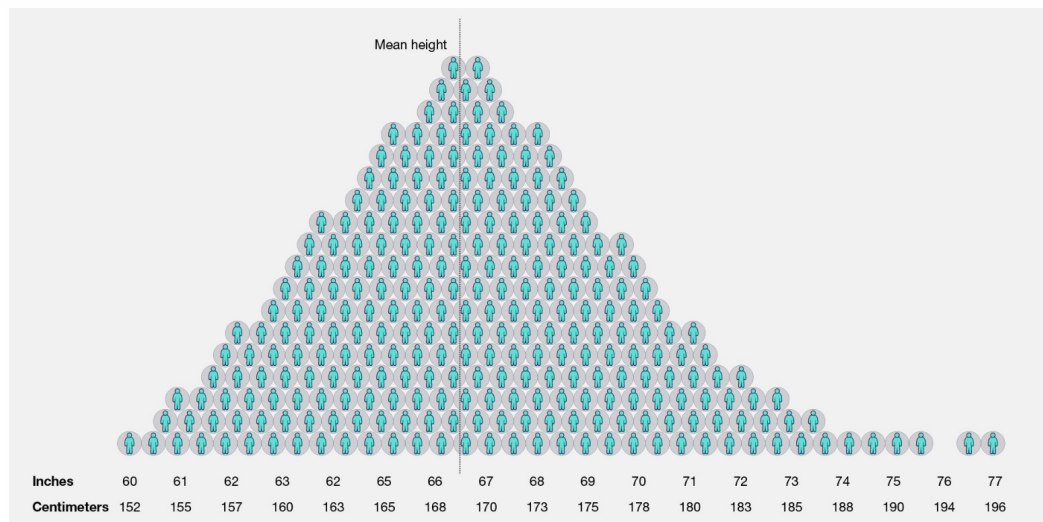


The **Central Limit Theorem** is a statistical principle stating that, under certain conditions, the distribution of the sum of a large number of independent and identically distributed random variables tends to approach a normal distribution, regardless of the original distribution's shape.

When we look at a population, we can measure the PHENOTYPIC variance

(that phenotypic variance is usually the sum of both genetic and environmental variances)

Phenotypic variance is the observed variability of a particular trait. For many measurable traits (such as height, skin color, and blood pressure) phenotypic variation can be thought of as a distribution. The variation within that distribution centers on an average or mean. Phenotypic variance is often caused by a combination of genetic and environmental factors.

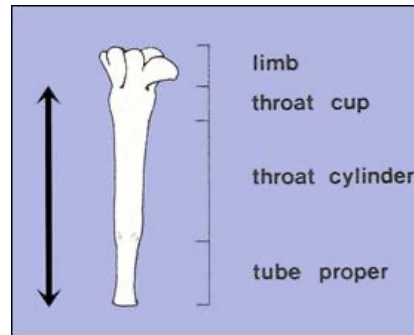


The variance of a complex trait:

An **experiment** to estimate the genetic and environmental variances

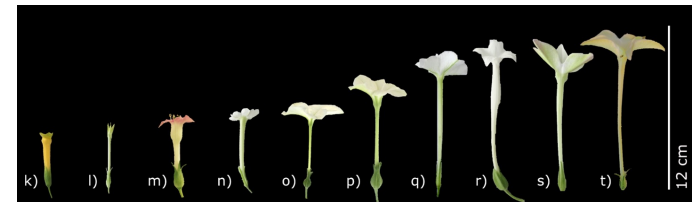
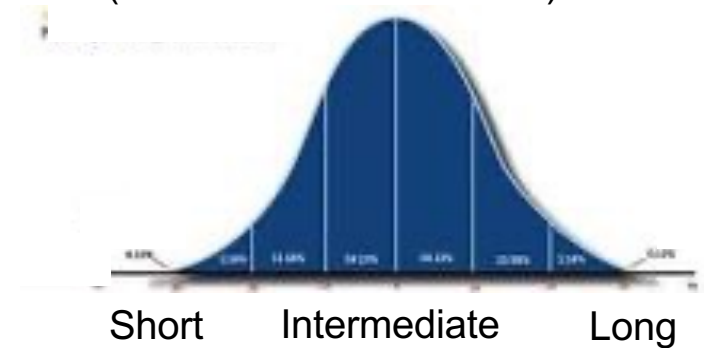


Tobacco

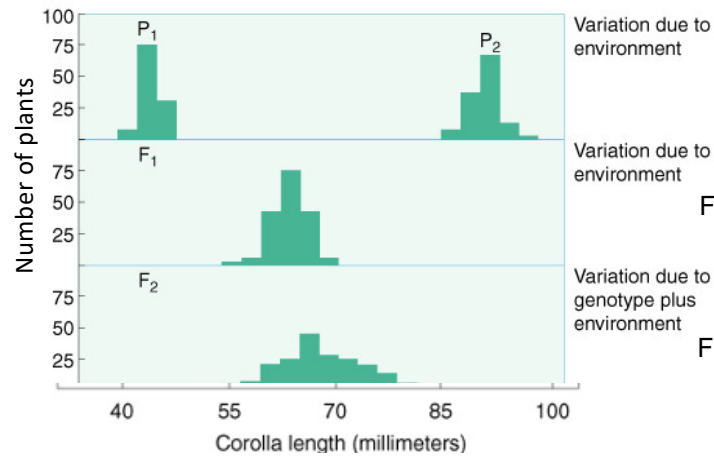


Complex trait=
Corolla length

Corolla length in wild tobacco:
(a **continuous** distribution)



Tobacco corolla length phenotypic variance (V_x)



P₁ and P₂ are **different inbred lines**
 (genetically homozygous... at some loci
they will be homozygous for different alleles)

F₁ **hybrid** is heterozygous for alleles that differ in P₁ and P₂.

F₂ progeny **segregate** for alleles that differ in P₁ and P₂.



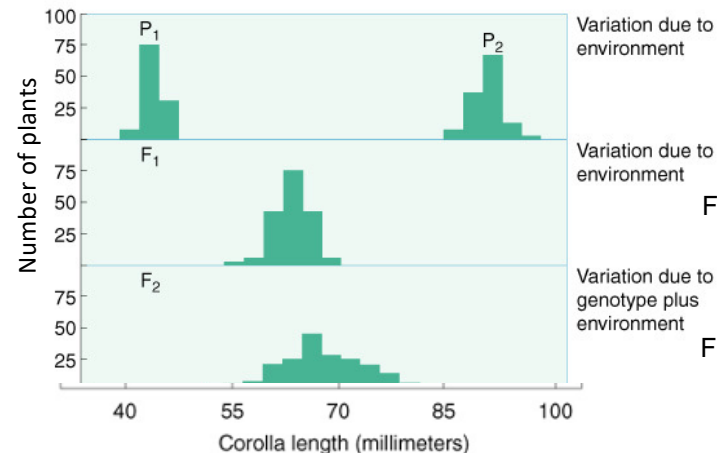
Tobacco corolla length phenotypic variance (V_x)

Source(s) of variance

Parents: Environmental differences between individuals

F_1 : Environmental differences between individuals

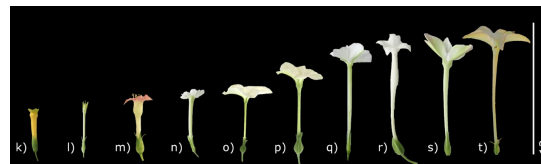
F_2 : Environmental differences between individuals
AND
 genetic differences between individuals



P₁ and P₂ are **different inbred lines**
 (genetically homozygous...though at some loci they will be homozygous for different alleles)

F₁ **hybrid** is heterozygous for alleles that differ in P₁ and P₂.

F₂ progeny **segregate** for alleles that differ in P₁ and P₂.



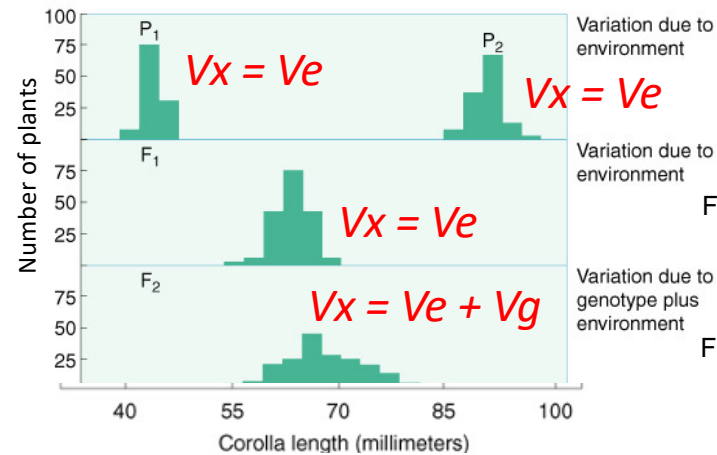
Tobacco corolla length phenotypic variance (V_x)

Source(s) of variance

Parents: Environmental differences between individuals

F₁: Environmental differences between individuals

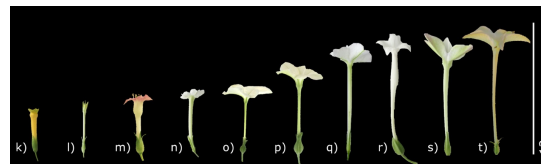
F₂: Environmental differences between individuals
AND
 genetic differences between individuals



P₁ and P₂ are **different inbred lines**
 (genetically homozygous...though at some loci they will be homozygous for different alleles)

F₁ **hybrid** is heterozygous for alleles that differ in P₁ and P₂.

F₂ **progeny** segregate for alleles that differ in P₁ and P₂.



Heritability

broad sense

$$H^2 = \frac{V_g}{V_X}$$

- Bounded by 0 and 1
- Dimensionless
- Standardized parameter
- Comparable across traits / species

You have a population of genetically identical mice of strain A. You also have a second population of genetically identical mice of strain B. The mean and variance of body weight of the strain A population are 30 g and 400 squared grams, respectively, while the mean and variance of body weight of strain B are 10 g and 400 squared grams, respectively.

You cross the strains to obtain an F1, which you allow to *interbreed*, and then raise the F2 progeny. The mean and variance of body weight of the F2 population is 20 g and 1200 squared grams. **What is the broad-sense heritability of body weight?**

- a. 0.67
- b. 0.50
- c. 0.33
- d. 0.25

Heritability

broad sense

$$H^2 = \frac{V_g}{V_X}$$

- Bounded by 0 and 1
- Dimensionless
- Standardized parameter
- Comparable across traits / species

Heritability Estimation in Humans: Twin Studies



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Twin 1 Twin 2



Twin Sisters Separated at Birth Reunite

[Visit](#)

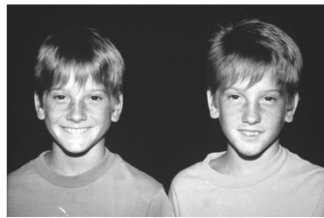
Twin 1 Twin 2

Table 19-4 Broad-Sense Heritability for Some Traits in Humans as Determined by Twin Studies

Trait	H^2
Physical attributes	
Height	0.88
Chest circumference	0.61
Waist circumference	0.25
Fingerprint ridge count	0.97
Systolic blood pressure	0.64
Heart rate	0.49
Mental attributes	
IQ	0.69
Speed of spatial processing	0.36
Speed of information acquisition	0.20
Speed of information processing	0.56
Personality attributes	
Extraversion	0.54
Conscientiousness	0.49
Neuroticism	0.48
Positive emotionality	0.50
Antisocial behavior in adults	0.41
Psychiatric disorders	
Autism	0.90
Schizophrenia	0.80
Major depression	0.37
Anxiety disorder	0.30
Alcoholism	0.50–0.60
Beliefs and political attitudes	
Religiosity among adults	0.30–0.45
Conservatism among adults	0.45–0.65
Views of school prayer	0.41
Views on pacifism	0.38

Sources: J. R. Alford et al., *American Political Science Review* 99, 2005, 1–15; T. Bouchard et al., *Science* 250, 1990, 223–228; T. Bouchard, *Curr. Dir. Psych. Sci.* 13, 2004, 148–151; P. J. Clark, *Am. J. Hum. Genet.* 7, 1956, 49–54; C. M. Freitag, *Mol. Psychiatry* 12, 2007, 2–22.

MZ and DZ twins



Identical



Fraternal: Same-Sex



Fraternal: Opposite-Sex

FIGURE 1. The study of twins. Photo credit: Nancy Segal.

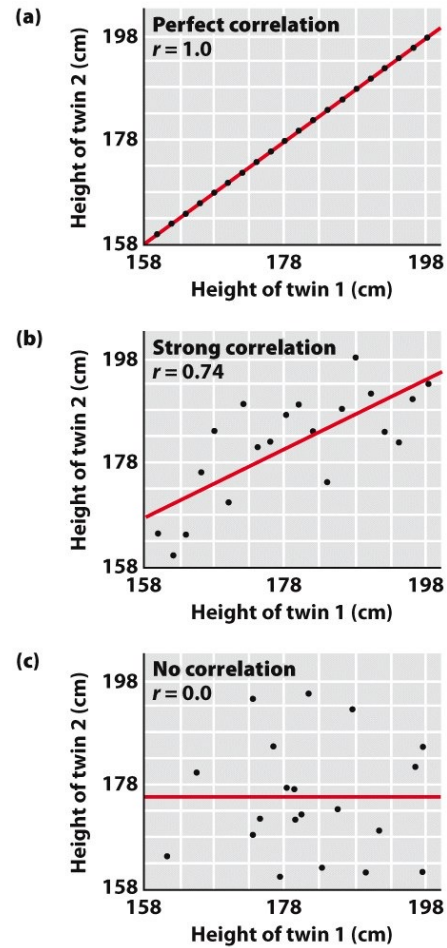


Figure 19-4
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Twin Sisters Separated at Birth Reunite

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TWIN STUDIES

Correlation and Broad Sense Heritability

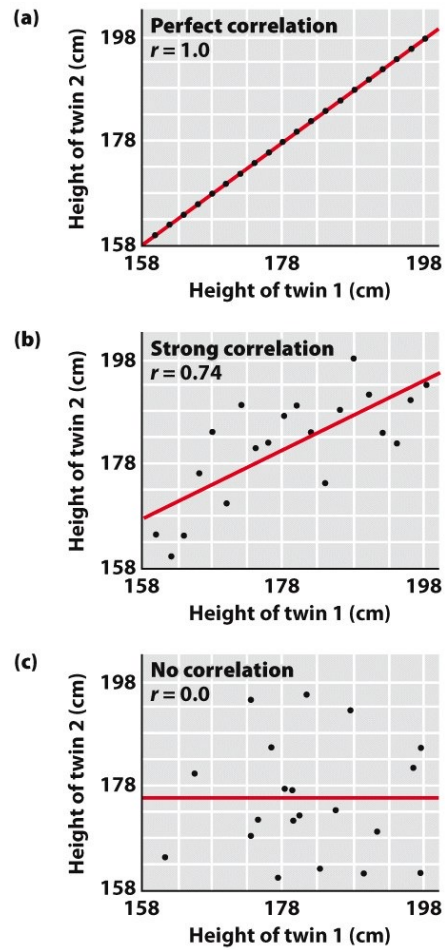
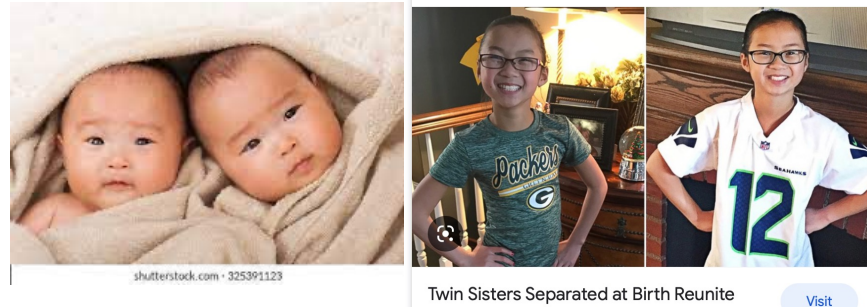


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TWIN STUDIES

Correlation and Broad Sense Heritability

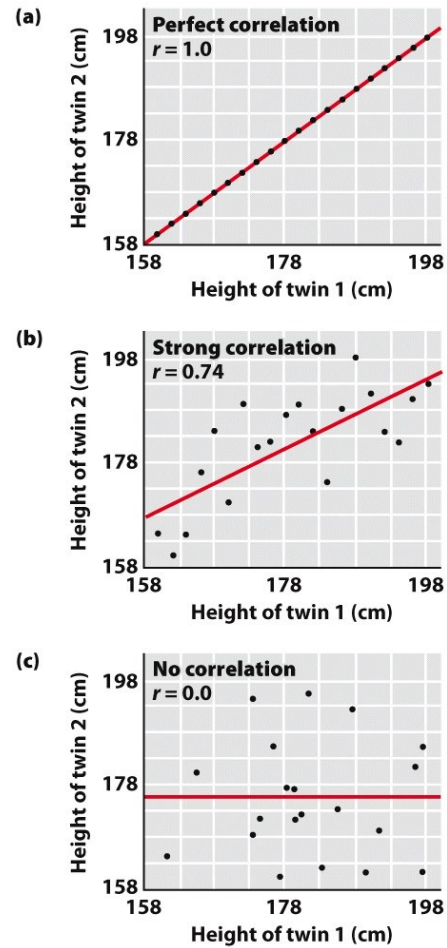
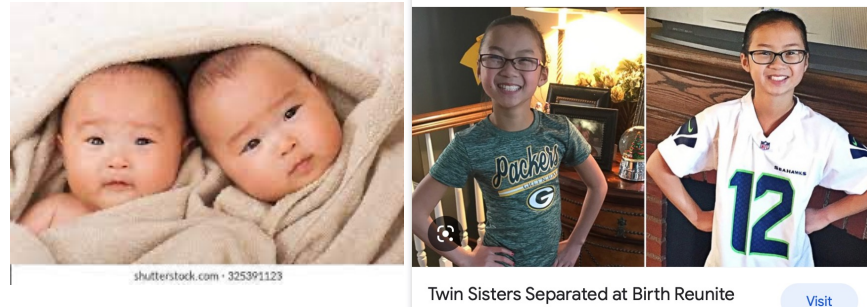


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TWIN STUDIES

Correlation and Broad Sense Heritability

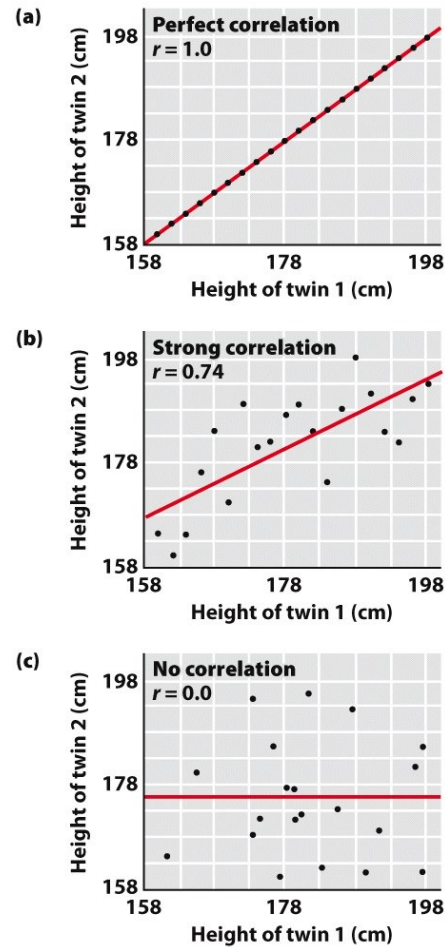
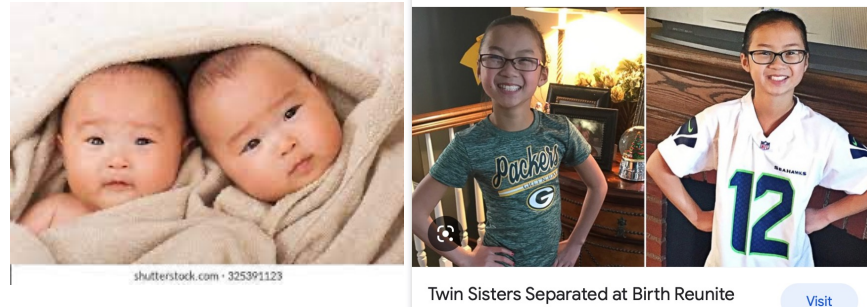


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TWIN STUDIES

Correlation and Broad Sense Heritability

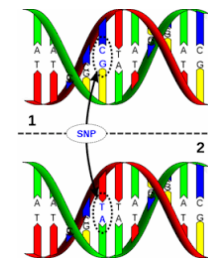
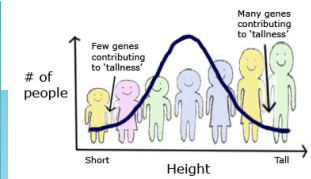
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Views on pacifism	0.38



Complex trait genomics

THE NEW FRONTIER

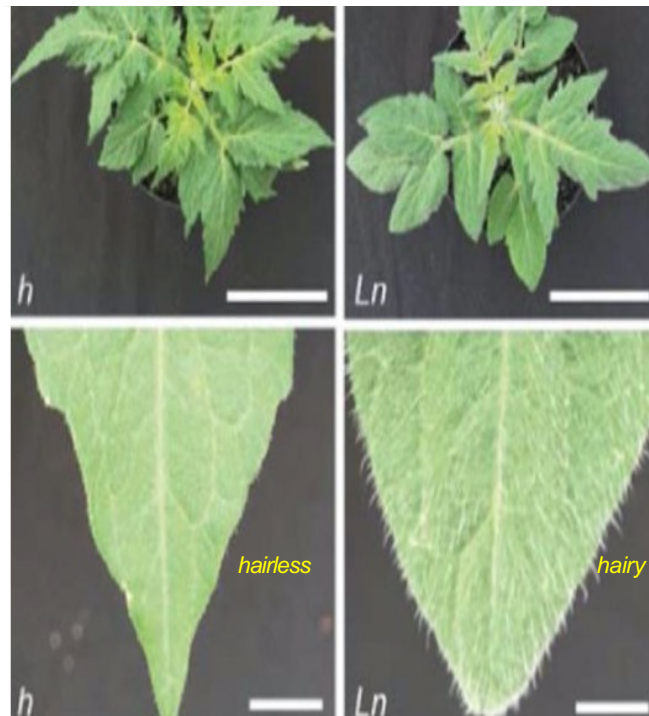


Using molecular markers to infer the presence and effects of genes that influence genetic variance (QTL and GWAS)



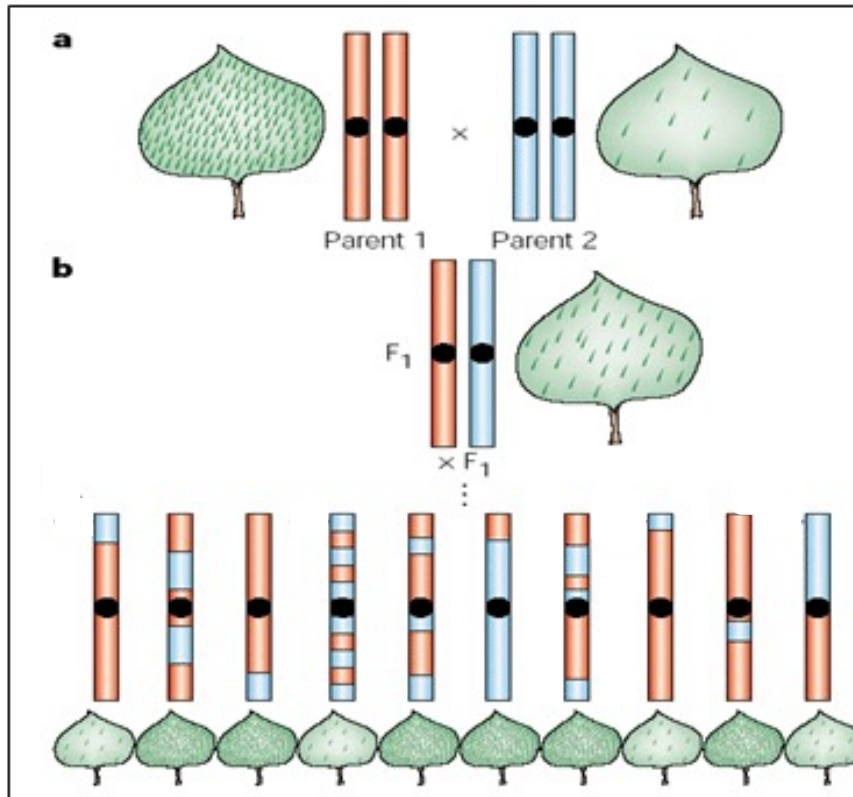
- **QTL mapping**...resembles classical mapping approaches (controlled crosses used)...**co-segregation of a marker with a trait's phenotypic** value helps you to locate the region of the genome where a gene with influence on the phenotype resides....there may be many such genes (called quantitative trait loci or QTL's)
- **Association mapping** or **genome-wide association studies (GWAS)**...relies on population surveys that look for correlation between markers and phenotypic trait values in samples taken from large populations.

Principles of QTL mapping: Co-segregation (co-inheritance) of phenotypes and marker alleles (*usually SNPs*)



Principles of QTL mapping: Co-segregation (co-inheritance) of phenotypes and marker alleles (*usually SNPs*)

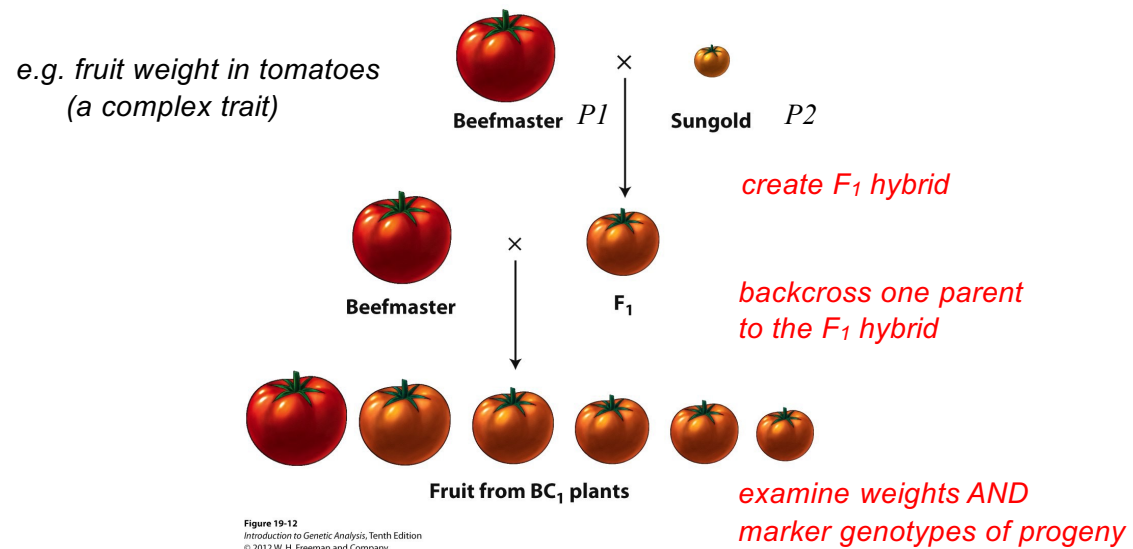
QTL mapping.....the basic approach:



Where is the gene that influences “hairiness”?

- A. At the bottom of the chromosome.
- B. In the middle of the chromosome.
- C. At the top of the chromosome.
- D. Can’t tell from the information.

QTL mapping begins with crosses of **two inbred lines**
(that differ in the trait of interest and their molecular marker genotypes)



QTL mapping

(Parents are homozygous for different "marker" alleles)

Markers (e.g., SNPs, Microsatellites, etc. on a **chromosome segment**)

Positions of markers

<i>M1</i>	<i>M2</i>	<i>M3</i>	<i>M4</i>	<i>M5</i>
-----------------	-----------------	-----------------	-----------------	-----------

Beefmaster alleles:

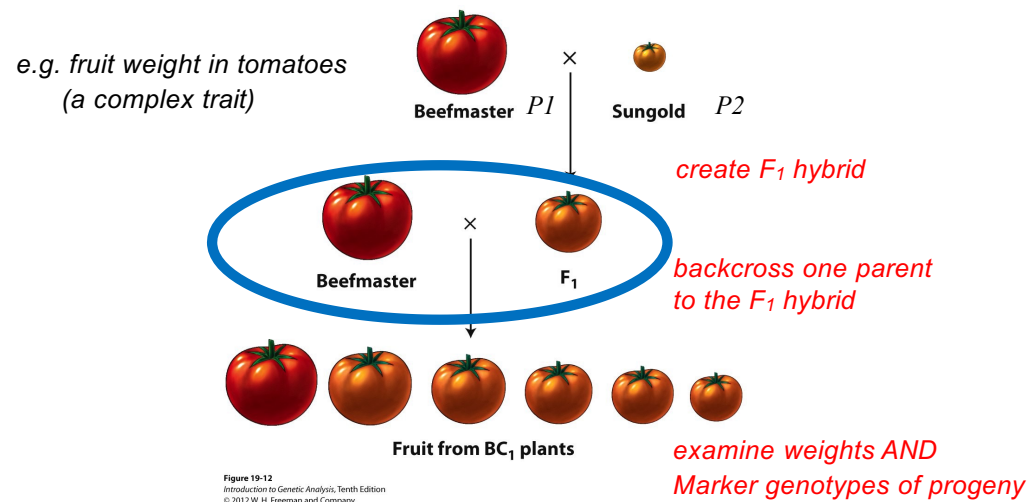
<i>B/B</i>	<i>B/B</i>	<i>B/B</i>	<i>B/B</i>	<i>B/B</i>
------------------	------------------	------------------	------------------	------------

Sungold alleles:

<i>S/S</i>	<i>S/S</i>	<i>S/S</i>	<i>S/S</i>	<i>S/S</i>
------------------	------------------	------------------	------------------	------------

Genes (QTLs) affecting the complex trait (fruit weight) could be anywhere in the chromosome segment...but **WHERE** are they and what are their effects?

QTL mapping begins with crosses of two inbred lines
(that differ in the trait of interest and their molecular marker genotypes)



QTL mapping (Recombination in F1 and crossovers)

Markers (e.g., SNPs, Microsatellites, etc.)

<i>M1</i>	<i>M2</i>	<i>M3</i>	<i>M4</i>	<i>M5</i>
-----------	-----------	-----------	-----------	-----------

F1

<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>
<i>S</i>	<i>S</i>	<i>S</i>	<i>S</i>	<i>S</i>

X

Beefmaster

<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>
<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>	<i>B</i>



Table 19-6 Simulated Fruit Weight and Marker-Locus Data for a Backcross Population between Two Tomato Inbred Lines—Beefmaster and Sungold

Plant	Fruit wt. (g)	Markers				
		M1	M2	M3	M4	M5
Beefmaster	230	B/B	B/B	B/B	B/B	B/B
Sungold	10	S/S	S/S	S/S	S/S	S/S
BC ₁ -001	183	B/B	B/B	B/B	B/S	B/S
BC ₁ -002	176	B/S	B/S	B/B	B/B	B/B
BC ₁ -003	170	B/B	B/S	B/S	B/S	B/S
BC ₁ -004	185	B/B	B/B	B/B	B/S	B/S
BC ₁ -005	182	B/B	B/B	B/B	B/B	B/B
BC ₁ -006	170	B/S	B/S	B/S	B/S	B/B
BC ₁ -007	170	B/B	B/S	B/S	B/S	B/S
BC ₁ -008	174	B/S	B/S	B/S	B/S	B/S
BC ₁ -009	171	B/S	B/S	B/S	B/B	B/B
BC ₁ -010	180	B/S	B/S	B/B	B/B	B/B
BC ₁ -011	185	B/S	B/B	B/B	B/S	B/S
BC ₁ -012	169	B/S	B/S	B/S	B/S	B/S
BC ₁ -013	165	B/B	B/B	B/S	B/S	B/S
BC ₁ -014	181	B/S	B/S	B/B	B/B	B/S
BC ₁ -015	169	B/S	B/S	B/S	B/B	B/B
BC ₁ -016	182	B/B	B/B	B/B	B/S	B/S
BC ₁ -017	179	B/S	B/S	B/B	B/B	B/B
BC ₁ -018	182	B/S	B/B	B/B	B/B	B/B
BC ₁ -019	168	B/S	B/S	B/S	B/B	B/B
BC ₁ -020	173	B/B	B/B	B/B	B/B	B/B
Mean of B/B	-	176.3	179.6	180.7	176.1	175.0
Mean of B/S	-	175.3	173.1	169.6	175.3	176.4
Overall mean	175.7					

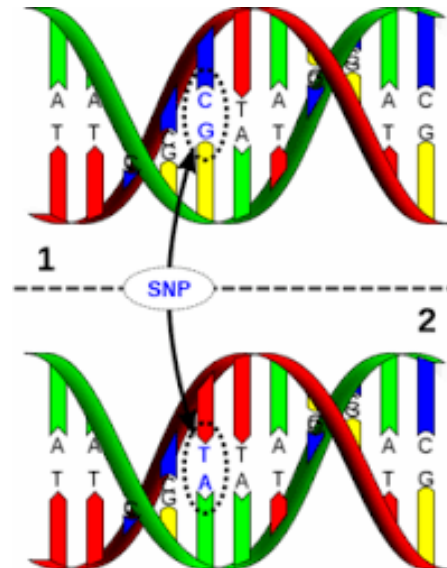
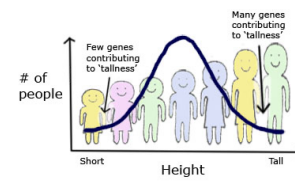
QTL mapping (Recombination in F1 and crossovers)

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BC ₁ -004	185	B/B	B/B	B/B	B/S	B/S
BC ₁ -005	182	B/B	B/B	B/B	B/B	B/B
BC ₁ -006	170	B/S	B/S	B/S	B/S	B/B
BC ₁ -007	170	B/B	B/S	B/S	B/S	B/S
BC ₁ -008	174	B/S	B/S	B/S	B/S	B/S
BC ₁ -009	171	B/S	B/S	B/S	B/B	B/B
BC ₁ -010	180	B/S	B/S	B/B	B/B	B/B
BC ₁ -011	185	B/S	B/B	B/B	B/S	B/S
BC ₁ -012	169	B/S	B/S	B/S	B/S	B/S
BC ₁ -013	165	B/B	B/B	B/S	B/S	B/S
BC ₁ -014	181	B/S	B/S	B/B	B/B	B/S
BC ₁ -015	169	B/S	B/S	B/S	B/B	B/B
BC ₁ -016	182	B/B	B/B	B/B	B/S	B/S
BC ₁ -017	179	B/S	B/S	B/B	B/B	B/B
BC ₁ -018	182	B/S	B/B	B/B	B/B	B/B
BC ₁ -019	168	B/S	B/S	B/S	B/B	B/B
BC ₁ -020	173	B/B	B/B	B/B	B/B	B/B
Mean of B/B	-	176.3	179.6	180.7	176.1	175.0
Mean of B/S	-	175.3	173.1	169.6	175.3	176.4
Overall mean	175.7					

Where is the gene that influences tomato size?

- A. Near SNP1
- B. Near SNP2
- C. Near SNP3
- D. Near SNP4
- E. Near SNP5



Main points—Complex traits

- Complex traits show continuous variation or liability
- Many genes and many environmental effects affect complex phenotypes
- We are interested in quantifying how much of the overall phenotypic variation for the trait is due to variation in the genetic make-up of individuals (the **heritability** of the trait)
- QTL analysis allow us to map genes that influence complex trait variation, and to assess their effects.